A twenty-six-variable consistency equation

Positive integer unknowns; exact degree 12; a 124-operation evaluator.

The only unknowns are the twenty-six distinct positive integers

$$J,F,\alpha,f,g,T,X,Y,\xi,\eta,\zeta,\upsilon,\gamma,\vartheta,\psi,\varphi,\epsilon,c,a,\ell,h,t,s,x,m,n\in\mathbb{Z}_{>0}.$$

All other symbols below are polynomial abbreviations:

$$\begin{gathered}
b=\beta J,\quad q=1+b,\quad C=q-F-\alpha,\quad U=Yq^2,\\
E=XY,\quad k=\eta+\zeta,\quad W=h-t,\\
H=3+4a,\quad \Delta=H+a^2,\quad L=X+ac,\\
M=sb,\\
R=[\alpha+(q-F)b](q^2-1)+J(\mu+\omega q),\\
Z=\psi(\ell+k+\xi-kW)+Yb-Ws,\\
D=\varphi L+H(\varphi\gamma+x+\epsilon+1-k),\\
I=ng,\quad Q=\Delta(m-1),\quad V=m(Tc-\epsilon)-c,\\
\mathcal{P}=Z^2-\psi^2[1+\ell(\ell+k)],\\
\mathcal{D}=D^2-\varphi^2(1+\Delta n),\\
\mathcal{Q}=\upsilon^2-Q[\upsilon^2-\vartheta]-1,\\
\mathcal{R}=I^2-Q,\quad \mathcal{E}=\vartheta-V^2,\\
\mathcal{B}=\psi-M,\quad \mathcal{C}=\varphi-U,\quad \mathcal{G}=\epsilon-R,\\
\mathcal{H}=c-\eta-Yk,\quad \mathcal{T}=a-E-Y,\\
\mathcal{L}=\ell-EYk,\quad \mathcal{W}=W-[C(X+Kq)-Fq],\\
\mathcal{X}=s-q^3,\quad \mathcal{Y}=x-Eq,\\
\mathcal{Z}=m-f^2,\quad \mathcal{N}=n-c^2.
\end{gathered}$$

The single Diophantine equation is

$$\begin{aligned}
&\mathcal{P}^2+\mathcal{D}^2+\mathcal{Q}^2+\mathcal{R}^2+\mathcal{E}^2+\mathcal{B}^2+\mathcal{C}^2+\mathcal{G}^2\\
&+\mathcal{H}^2+\mathcal{T}^2+\mathcal{L}^2+\mathcal{W}^2+\mathcal{X}^2+\mathcal{Y}^2+\mathcal{Z}^2+\mathcal{N}^2=0.
\end{aligned}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho=2^{47},\quad B=\rho^{2011},\quad \beta=B-1,\\
G_m(x)=\sum_{j=0}^{m-1}x^j,\quad \Lambda=\rho^{1005}.
\end{gathered}$$

$$\begin{aligned}
\mu&=\beta-G_{951}(\rho)\\
&\quad-\rho^{960}G_{26}(\rho)-\rho^{1004},\\
\omega&=\beta+8G_{917}(\rho)-4+(2^{45}-4)\Lambda\\
&\quad+\rho^{960}G_9(\rho)+\rho^{987}G_8(\rho)+\rho^{1004},\\
K&=\rho^{18}+\rho^{1004}+\rho^{2020}\\
&\quad+2\Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{gathered}
\chi_i=1+2^{10}(1+w_i+2^8n_i\\
+2^{17}p_i+2^{25}s_i)\quad(0\le i\le 941),\\
\pi(0)=0,\quad \pi(i)=1+\sum_{j=1}^{941}\left\lfloor\frac{2^{44}-1+\chi_j-\chi_i}{2^{44}}\right\rfloor\quad(1\le i\le 941).
\end{gathered}$$

$$\begin{aligned}
\Phi(x)=&\sum_{i=0}^{941}\chi_i x^{\pi(i)}\\
&+2^{10}x^{969}(x^9+2^{17})G_8(2x)\\
&+2^{18}x^{942}(x^{18}+2^{17})G_9(2x).
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j,v_j,d_j decoded below. For $d=0,1$, let

$$\begin{gathered}
h(q,b)=2q+b+2,\quad \mathcal{S}_d=\{u_j:d_j=d,\ 0\le j<430\},\\
r_{d,q}=205-199d+|\mathcal{S}_d\cap\{0,\dots,q-1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0\le i\le 941$; all index ranges are fixed:

index i	tile $t_i=(w_i,n_i,p_i,s_i)$
0	$(2,435,3,1)$
$j+1$ $(j=2q+b<430)$	$((1-d_j)r_{0,u_j},h(q,b),d_jr_{1,u_j},v_j)$
$433-2a+b$ $(a,b=0,1)$	$(0,432a+b,0,432a+b)$
$\substack{423+2r_{d,q}+b\\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q},b,(1-d)r_{d,q},h(q,b))$
$923+2a+b$ $(a,b=0,1)$	$(0,432a+b,0,438)$
927	$(2,434,2,0)$
928	$(3,436,4,4)$
929	$(4,436,4,0)$
$496+t$ $(t=434,435,436)$	$(1,438,1,t)$
$933+3a+b$ $(a,b=0,1,2)$	$(a^2,437,b,437)$

A closed decoder for the 430 actions. Put $D_b(A,k)=\left\lfloor A/b^k\right\rfloor \bmod b$, where $a \bmod b = a-b\lfloor a/b\rfloor$. For $0\le q<215$, $b=0,1$, $j=2q+b$:

$$\begin{gathered}
\mathcal{N}=\mathcal{A}\bmod 2^{636},\quad \mathcal{T}=\left\lfloor\mathcal{A}/2^{636}\right\rfloor,\\
\nu_j=D_2(\mathcal{N},j),\quad \kappa_j=(\sigma_j+2)^{1-\nu_j}\quad(0\le j<430),\\
\sigma_j=\sum_{k=0}^{j-1}\nu_k,\quad \Pi_j=\prod_{k=0}^{j-1}\kappa_k\quad(0\le j\le 430),\\
m_q=D_2(\mathcal{N},430+q),\quad \tau_q=\sum_{k=0}^{q-1}m_k,\\
F_q=14+m_q(D_{16}(\lfloor\mathcal{T}/\Pi_{430}\rfloor,\tau_q)-14),\\
u_j=(\sigma_{j+1}-(\lfloor\mathcal{T}/\Pi_j\rfloor \bmod \kappa_j)) \bmod 216,\\
v_j=D_2(F_q,2b),\quad d_j=D_2(F_q,2b+1).
\end{gathered}$$

Empty sums/products are 0/1. Here $(v_0,d_0,u_0)=(1,1,1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

```
12889260410359561746736540743515845134244260640012702033212621
95398074180784058571518488282119861023093355389406939414042892
97455773992193784795705001618777025966369137954859764532832722
75411843410809038713502633204617333246733616940964578272360901
56186216508504953649192752154519268024509952082950238865468307
22999632482769253980274091217460707829023970817244790340704449
09085581271192071656170732004937337082171141082486384269072884
72778087419663417942892520175630649792329265834490253035577533
96449689808375788457488971017043689875433162972390177936123032
89025657757489526342497438772045714099661532481123050235898973
27443609903800541199351405985754278307227777584454508422717998
33411356440933534607489268032843 7
```

The fixed integer $\mathcal{A}$ above is written in base 10. Lean proves that this equation has a positive solution exactly when the twenty-two-variable degree-16 equation has one, on the documented fixed source snapshot. The native intuitionistic PRA endpoint for this equation is being checked. The exact degree is certified by integer-polynomial calculations; the 124-operation count is for the unshifted circuit. Neither count is a minimality claim.

Exact 124-operation circuit

Each numbered line defines one gate over the integers. The inputs are the twenty-six unknowns and the four fixed integers above. The ASCII names `alpha, xi, eta, zeta, upsilon, gamma, vartheta, psi, varphi, epsilon, beta, mu, omega, ell, Delta` refer to their displayed counterparts; `g` is the unknown g. The last gate is zero exactly when the displayed equation holds. The circuit inputs `cNamed, aNamed, ellNamed` are c, a, ℓ; `WNamed` is the displayed W, while gate `W` is $C(X + Kq) - Fq$. The inputs `s, x, m, n` are the displayed s, x, m, n. Intermediate gate names below are local abbreviations, not additional unknowns.

```
1.  b = J * beta
2.  q = 1 + b
3.  q2 = q * q
4.  q3 = q * q2
5.  U = Y * q2
6.  qF = q - F
7.  C = qF - alpha
8.  E = X * Y
9.  k = eta + zeta
10.  kY = Y * k
11.  c = eta + kY
12.  c2 = cNamed * cNamed
13.  a = E + Y
14.  g15 = 4 * aNamed
15.  H = 3 + g15
16.  g17 = aNamed * aNamed
17.  Delta = H + g17
18.  g19 = aNamed * cNamed
19.  L = X + g19
20.  ell = E * kY
21.  M = s * b
22.  g25 = K * q
23.  g26 = X + g25
24.  g27 = C * g26
25.  g28 = F * q
26.  W = g27 - g28
27.  g30 = ellNamed + k
28.  g31 = g30 + xi
29.  WNamed = h - t
30.  root_skew = k * WNamed
31.  root_inner = g31 - root_skew
32.  root_scaled = psi * root_inner
33.  g33 = Y * b
34.  root_btq3 = WNamed * s
35.  root_sum = root_scaled + g33
36.  Z = root_sum - root_btq3
37.  g38 = Z * Z
38.  g39 = psi * psi
39.  g40 = ellNamed * g30
40.  g41 = 1 + g40
41.  g42 = g39 * g41
42.  A0 = g38 - g42
43.  g44 = qF * b
44.  rankseed = alpha + g44
45.  g46 = q2 - 1
46.  g47 = rankseed * g46
47.  g48 = omega * q
48.  g49 = mu + g48
49.  g50 = J * g49
50.  R = g47 + g50
51.  g52 = k - epsilon
52.  r0 = g52 - 1
53.  g54 = varphi * gamma
54.  g55 = E * q
55.  g56 = g54 + x
56.  shift = g56 - r0
57.  g58 = varphi * L
58.  g59 = H * shift
59.  D = g58 + g59
60.  g61 = D * D
61.  g62 = varphi * varphi
62.  g63 = Delta * n
63.  g64 = 1 + g63
64.  g65 = g62 * g64
65.  A1 = g61 - g65
66.  f2 = f * f
67.  rho = m - 1
68.  Q = Delta * rho
69.  I = n * g
70.  g71 = I * I
71.  g72 = g71 - Q
72.  g74 = T * cNamed
73.  g75 = g74 - epsilon
74.  g76 = m * g75
75.  V = g76 - cNamed
76.  y2 = upsilon * upsilon
77.  g80 = V * V
78.  g81 = y2 - vartheta
79.  g82 = Q * g81
80.  Fa = y2 - g82
81.  auxResidual = Fa - 1
82.  firstSquare = A0 * A0
83.  mainSquare = A1 * A1
84.  auxSquare = auxResidual * auxResidual
85.  strongSquare = g72 * g72
86.  firstMain = firstSquare + mainSquare
87.  firstMainAux = firstMain + auxSquare
88.  sumSquares = firstMainAux + strongSquare
89.  difference = vartheta - g80
90.  constraint = difference * difference
91.  result = sumSquares + constraint
92.  scaleDiff = psi - M
93.  scaleCheck = scaleDiff * scaleDiff
94.  final = result + scaleCheck
95.  mainDiff = varphi - U
96.  mainCheck = mainDiff * mainDiff
97.  result16 = final + mainCheck
98.  rankDiff = epsilon - R
99.  rankCheck = rankDiff * rankDiff
100.  result17 = result16 + rankCheck
101.  cDifference = cNamed - c
102.  cConstraint = cDifference * cDifference
103.  aDifference = aNamed - a
104.  aConstraint = aDifference * aDifference
105.  ellDifference = ellNamed - ell
106.  ellConstraint = ellDifference *
ellDifference
107.  wDifference = WNamed - W
108.  wConstraint = wDifference * wDifference
109.  addC = result17 + cConstraint
110.  addA = addC + aConstraint
111.  addEll = addA + ellConstraint
112.  finalWithNamedValues = addEll +
wConstraint
113.  cubeDifference = s - q3
114.  cubeConstraint = cubeDifference *
cubeDifference
115.  productDifference = x - g55
116.  productConstraint = productDifference *
productDifference
117.  fSquareDifference = m - f2
118.  fSquareConstraint = fSquareDifference *
fSquareDifference
119.  cSquareDifference = n - c2
120.  cSquareConstraint = cSquareDifference *
cSquareDifference
121.  addCube = finalWithNamedValues +
cubeConstraint
122.  addProduct = addCube + productConstraint
123.  addFSquare = addProduct +
fSquareConstraint
124.  addCSquare = addFSquare +
cSquareConstraint
```